

INDU 211: Introduction to Production & Manufacturing Systems

Master Study Guide & Exam Strategy Guide

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1. Welcome to INDU 211 — The Engineer's Big Picture

If this is your first course in Industrial Engineering, welcome! Most engineering disciplines focus heavily on the physical object itself:

- **Mechanical Engineers** design moving mechanisms (gears, linkages, thermal cycles).
- **Civil Engineers** design stationary structural systems (bridges, foundations, skyscrapers).
- **Electrical Engineers** design circuits, electromagnetic paths, and signal processors.

Industrial Engineers (IEs) do something fundamentally different: **We design, analyze, and optimize the *entire system* that brings ideas into reality.** That system includes machines, materials, money, information, and most importantly—**people**.

The IE Core Philosophy:

*"Work with **PEOPLE** to make systems **BETTER, FASTER, SAFER, and CHEAPER.**"*

2. The 5-Pillar Architecture of INDU 211

To succeed in INDU 211 without feeling overwhelmed by lecture slides, organize your mental model around five core pillars:

Pillar 1: Identity & Ethics	Pillar 2: Systems & Decisions	Pillar 3: Process & Economics	Pillar 4: Industrial Processes	Pillar 5: Facilities & Logistics
• Latin root: <i>Ingenium</i> • Science vs. Engineering • OIQ & Engineers Canada • Public safety vs. costs • IE Pioneers (Babbage to AI)	• Open vs. Closed Loops • Conversion Process Model • Strategic, Tactical, Control • Human Activity Systems • Management Systems	• Design vs. Production • Concurrent Engineering • Bill of Materials (BOM) • Fixed vs. Variable Costs • Break-Even & Crossover	• Refining & Alloying • Sand vs. Die Casting • Hot vs. Cold Forming • Machining (Turn/Mill/Drill) • Jigs vs. Fixtures & Tooling	• Macro Location vs. Layout • Euclidean vs. Rectilinear • Center of Gravity & Medians • Transportation Problem • 5 Layout Configurations

3. Master Glossary of High-Frequency Terms

Mastering INDU 211 exams requires fluent recall of exact terminology. Memorize these core concepts:

Term	Exact Definition	Key Exam Identifier
Ingenium	Latin root of <i>engineer</i> and <i>ingenious</i> ; means talent, natural capacity, or clever invention.	Etymology of engineering.
Science	Systematic quest for basic knowledge through observation, hypothesis, and controlled testing.	Goal: <i>Understanding / Theories</i> .
Engineering	Application of scientific and mathematical knowledge to create solutions for human welfare.	Goal: <i>Better life / Functional Systems</i> .
Analysis	Resolving or breaking down an existing system into its fundamental components.	Breaking down an <i>existing</i> system.
Synthesis	Combining disparate elements and concepts together to form a novel, unified whole.	Creating a <i>new</i> system.
OIQ	<i>Ordre des ingénieurs du Québec</i> ; regulatory body overseeing ~55,000 professional engineers in Quebec.	Monitors ethics, public safety, and professional practice.
Engineers Canada	National federation of the 12 provincial and territorial engineering regulators.	National standards and practice.
CEAB	<i>Canadian Engineering Accreditation Board</i> ; accredits undergraduate Canadian engineering degrees.	Validates academic qualifications for licensing.
Practice of Engineering	Any act of planning, designing, evaluating, advising, supervising requiring engineering principles.	Legal definition; public safety priority.
Open-Loop System	A system that executes an action without monitoring its output or course-correcting.	"Car without a driver"; no feedback.
Closed-Loop System	A system that measures its output, compares it to a goal, and applies feedback adjustments.	"Car with a driver"; feedback-driven.
Strategic Decision	Long-term, high-stakes executive choices: <i>What</i> to make, <i>How</i> to make it, <i>Where</i> to build.	3–10 year horizon; plant location, product line.
Tactical Decision	Medium-term operational resource allocation: <i>How much</i> to produce and <i>When</i> .	Monthly/quarterly; aggregate planning, inventory.
Control Decision	Immediate, day-to-day management of shop-floor activities and line workflows.	Daily/hourly; shift scheduling, machine dispatching.
Concurrent Engineering	Simultaneous integration of design, manufacturing, testing, maintenance, and recycling from Day 1.	Eliminates "throwing design over the wall".
Tolerance	Total allowable variation in a physical dimension from its nominal specification.	Tighter tolerance = exponentially higher cost.
Bill of Materials (BOM)	Hierarchical product structure tree listing all assemblies, components, quantities, and raw materials.	Multi-level parts list / recipe.
Fixed Cost (FC)	Capital expenditures that do not vary with production output (machines, plant footprint, tooling).	b in $Y = aX + b$.

Term	Exact Definition	Key Exam Identifier
Variable Cost (VC)	Per-unit expenses directly tied to volume (raw materials, direct labor, tool wear, consumable power).	a in $Y = aX + b$.
Break-Even Point (BEP)	Production/sales volume where total revenue equals total costs ($TR = TC$), resulting in zero net profit.	$Q_{BEP} = \frac{FC}{P-v}$.
Process Crossover Volume	Output volume where two competing manufacturing processes yield identical total production costs.	$FC_1 + v_1Q = FC_2 + v_2Q$.
Hot Working	Plastic deformation performed <i>above</i> metal recrystallization temperature; low forces, rough surface.	No work hardening; high ductility.
Cold Working	Plastic deformation performed <i>below</i> recrystallization temperature; high forces, strain hardening.	Superior surface finish & tight tolerances.
Casting	Pouring molten metal into a hollow mold cavity where it solidifies to assume the shape.	Near-net-shape for complex geometries.
Extrusion	Forcing billet metal through a shaped die orifice under high compressive pressure.	Constant cross-section profiles (pipes, rails).
Forging	Shaping metal using localized compressive impact forces (hammering or pressing).	Exceptional grain flow, toughness & strength.
Turning	Machining operation where workpiece <i>rotates</i> against a stationary single-point cutting tool (Lathe).	Generates cylindrical geometries.
Milling	Workpiece feeds against a rapidly <i>rotating multi-tooth cutter</i> (Milling machine).	Generates flat faces, slots, contours.
Shaping	Reciprocating machining where workpiece is stationary and cutting tool reciprocates.	Low-volume flat surfacing.
Planing	Reciprocating machining where tool is stationary and heavy workpiece moves beneath it.	Large workpieces (rail tracks, large beds).
Fixture	Rigid production device that locates and securely clamps a workpiece in a fixed coordinate frame.	Holds & locates workpiece (e.g. vise, clamp).
Jig	Production device that locates/holds workpiece AND directly guides the cutting tool.	Holds + Locates + Guides Tool (e.g. drill bushing).
Euclidean Distance (L_2)	Geometric straight-line distance: $d_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.	"As the crow flies"; intercity transit, pipelines.
Rectilinear Distance (L_1)	Orthogonal travel constrained to a grid: $d_R = \ x_1 - x_2\ + \ y_1 - y_2\ $.	Urban city blocks, factory floor aisles, AGVs.
Center of Gravity (Centroid)	Volume-weighted average coordinate minimizing squared Euclidean distance: $\bar{x} = \frac{\sum Q_i x_i}{\sum Q_i}$.	Siting central warehouses, distribution hubs.
Rectilinear 1-Median	Location minimizing weighted rectilinear distance; found at the coordinate-wise 50% cumulative volume point.	Optimal facility site in grid cities / plants.

Term	Exact Definition	Key Exam Identifier
Transportation Problem	Linear program optimizing shipments from m sources to n destinations to minimize freight costs.	Multi-plant, multi-warehouse distribution.
Least-Cost Assignment	Greedy heuristic allocating maximum units to the active cell with the lowest unit freight cost.	Initial feasible solution for transportation model.
The 30% to 95% Rule	Material handling accounts for 30% to 95% of total manufacturing cost; adds no value to product.	Core economic justification for layout design.
Product Layout	Sequential equipment layout following progressive operation order (assembly lines).	High volume, low variety, smooth flow, low WIP.
Process Layout (Job Shop)	Functional grouping of identical equipment into departments (lathes together, mills together).	Low volume, high variety, high WIP, flexible.
Cellular Layout (GT)	Grouping dissimilar machines into U-cells dedicated to part families (Group Technology).	Mass customization, low setup, agile flow.
Fixed-Position Layout	Product remains stationary; labor, equipment, and materials travel to the product.	Ships, large commercial aircraft, rockets, bridges.
Mixed / Hybrid Layout	Integrating multiple layout types across production stages (Process → Cellular → Product).	Modern automotive and electronics manufacturing.

4. High-Yield Exam Question Archetypes & Winning Formats

Archetype 1: Definitional & Etymological Questions

- **Example Question:** "Define the practice of engineering and explain why licensing bodies like the OIQ exist."
- **Scoring Formula:**
 1. Give the legal definition keyword set (*planning, designing, composing, evaluating, supervising requiring engineering principles*).
 2. State the primary mandate: safeguarding **life, health, property, public welfare, and the environment**.
 3. Address the conflict of interest: Engineers are employed by private corporations under cost pressure, but legally and ethically bound to public safety over company profits.

Archetype 2: Comparative Distinction Questions

- **Example Question:** "Contrast a Jig from a Fixture. Give an industrial example of each."
- **Scoring Formula:**
 - Create a clean 3-row markdown table:
 - **Primary Function:** Fixture holds and locates; Jig holds, locates, AND mechanically guides the cutting tool.
 - **Complexity & Cost:** Jigs feature guide bushings or hardened guides; fixtures rely on machine kinematics.

- **Examples:** Fixture = milling vise or welding clamp fixture; Jig = drill jig with hardened drill bushings.

Archetype 3: Quantitative Process Selection / Break-Even Questions

- **Example Question:** "Given Process A ($FC = \$110k, v = \2), Process B ($FC = \$80k, v = \4), Process C ($FC = \$75k, v = \5), determine the optimal process for any demand Q ."
- **Scoring Formula:**
 1. Calculate pairwise crossover points algebraically:

$$Q_{B-C} = \frac{80,000 - 75,000}{5 - 4} = 5,000 \text{ units}, \quad Q_{A-B} = \frac{110,000 - 80,000}{4 - 2} = 15,000 \text{ units}$$

2. State the final selection intervals clearly:
 - $0 \leq Q < 5,000$: Select **Process C** (lowest fixed cost).
 - $5,000 < Q < 15,000$: Select **Process B** (balanced middle ground).
 - $Q > 15,000$: Select **Process A** (lowest variable cost dominates at scale).

Archetype 4: Operation Sequencing Rules

- **Example Question:** "State the fundamental rules for sequencing manufacturing operations on a raw steel shaft."
- **Scoring Formula:**
 1. **Shortest path:** Minimize material handling distance; eliminate backtracking.
 2. **Non-interference:** Ensure subsequent operations do not damage previously finished surfaces.
 3. **Datum preservation:** Group as many cuts as possible on a single machine/setup before releasing the workpiece.
 4. **Rough before finish:** Heavy material removal before heat treatment and final precision cylindrical grinding.

Archetype 5: Center of Gravity & Rectilinear Median Modeling

- **Example Question:** "Given regional destinations with coordinates (x_i, y_i) and demand volumes Q_i , calculate the optimal facility location under unweighted and weighted conditions. Explain how rectilinear distance differs."
- **Scoring Formula:**
 1. Show clean tabular calculation of $\sum Q_i x_i$ and $\sum Q_i y_i$.
 2. Compute centroid: $\bar{x} = \frac{\sum Q_i x_i}{\sum Q_i}, \bar{y} = \frac{\sum Q_i y_i}{\sum Q_i}$.
 3. Explain that Center of Gravity minimizes *squared Euclidean distance*.
 4. Contrast with Rectilinear travel ($d_R = |\Delta x| + |\Delta y|$), explaining that the true rectilinear optimum is found by the independent weighted 50% median rule.

Archetype 6: Transportation Tableau & Least-Cost Allocation

- **Example Question:** "Using the Least-Cost assignment method, find an initial feasible solution for a balanced transportation matrix and evaluate total cost."
- **Scoring Formula:**

1. Confirm supply-demand balance ($\sum S_i = \sum D_j$).
2. Greedily allocate maximum feasible units to the cell with lowest unit cost c_{ij} .
3. Cross off exhausted row or column, adjust residual supply/demand, and repeat.
4. Calculate total cost: $TC = \sum \sum c_{ij} x_{ij}$.
5. Note that alternative tie-break choices can yield higher costs, demonstrating the need for exact linear programming algorithms.

Archetype 7: Plant Layout Configuration Trade-Offs

- **Example Question:** "Contrast Product Layout and Process Layout across volume, WIP inventory, material handling expense, and line failure vulnerability."
- **Scoring Formula:**
 - Build a 4-point comparative breakdown:
 - **Volume/Variety:** Product = High vol/low variety; Process = Low vol/high variety.
 - **WIP Inventory:** Product = Minimal (continuous line); Process = Extremely high (parts wait in queues).
 - **Material Handling:** Product = Fixed conveyors (low cost/unit); Process = Forklifts/manual carts (high cost/unit; 30–95% rule).
 - **Line Stoppage:** Product = Catastrophic (1 machine stops entire plant); Process = Resilient (work is rerouted).

5. Course Timeline & Exam Study Strategy

Phase 1: Foundations (Weeks 1–2)	Phase 2: Manufacturing (Weeks 3)	Phase 3: Facilities & Layout (Weeks 4–5)	Phase 4: Operations & Midterm (Weeks 6+)
• Science vs. Engineering • OIQ Ethics & Regulations • Systems & Feedback Loops • Strategic/Tactical/Control • 8 IE Historical Pioneers	• Concurrent Engineering • Multi-level BOM Trees • Cost-Volume Break-Even • Metallurgy & Metal Forming • Machining, Cutting & Welding	• Macro Location Criteria • Euclidean vs. Rectilinear • Center of Gravity & Medians • Transportation Model • 5 Layout Configurations	• Master 1-Page Cheatsheets • Solve Multi-Process BEP • Solve Transportation Tableaus • Center of Gravity Drills • High-Yield Term Flashcards

1. **Deep Comprehension:** Read the [Comprehensive Topic Guides](#) to understand *why* each system behaves the way it does.
2. **Quantitative Muscle Memory:** Work through both quantitative guides in [04 - Worked Problems](#) for Cost-Volume crossover and Facilities Location/Transportation until you can solve them effortlessly.
3. **Rapid Recall:** Use the [1-Page Rapid Review Sheets](#) 48 hours before exams to lock in key distinctions (Jig vs Fixture, Product vs Process layout, Center of Gravity vs Median).